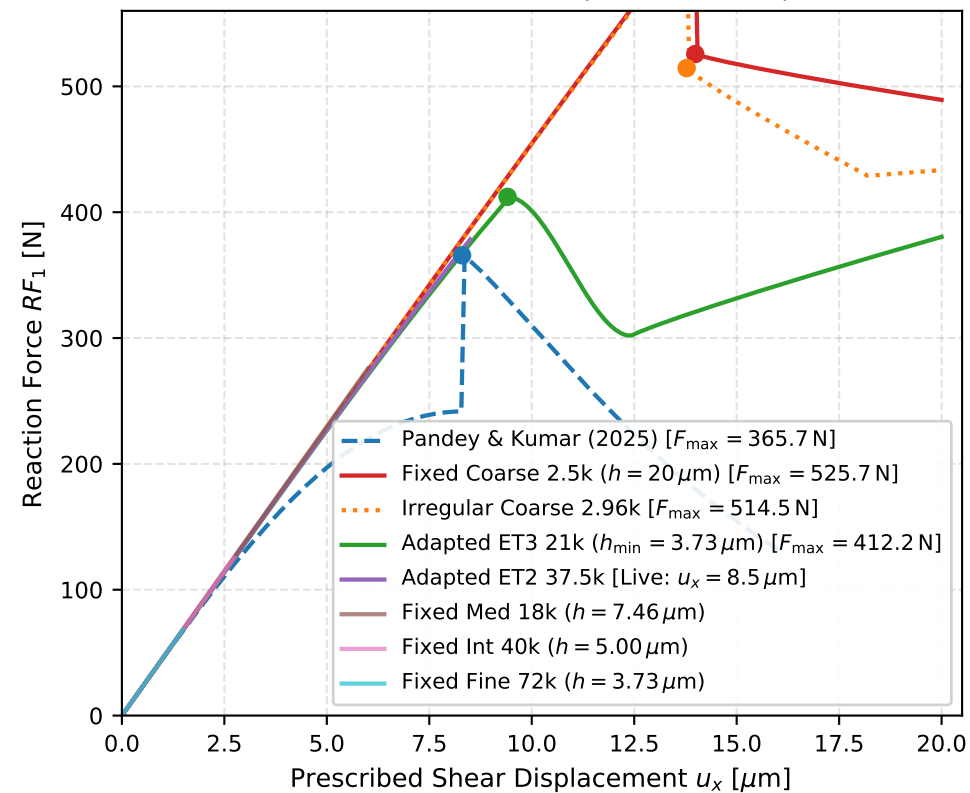
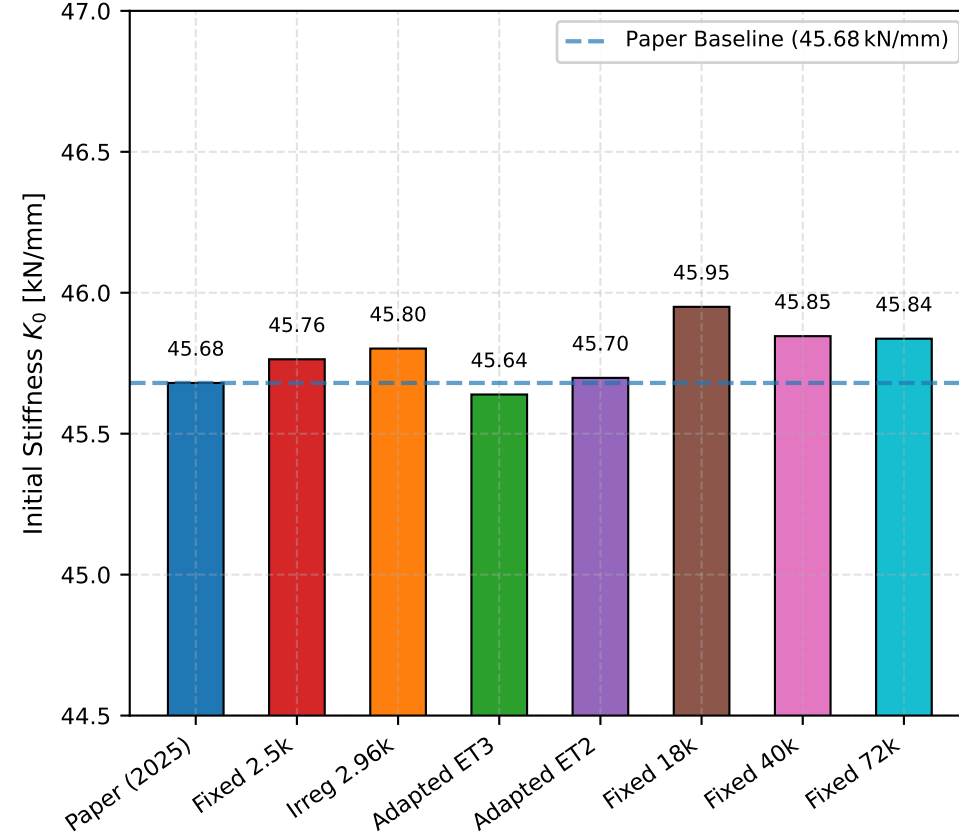
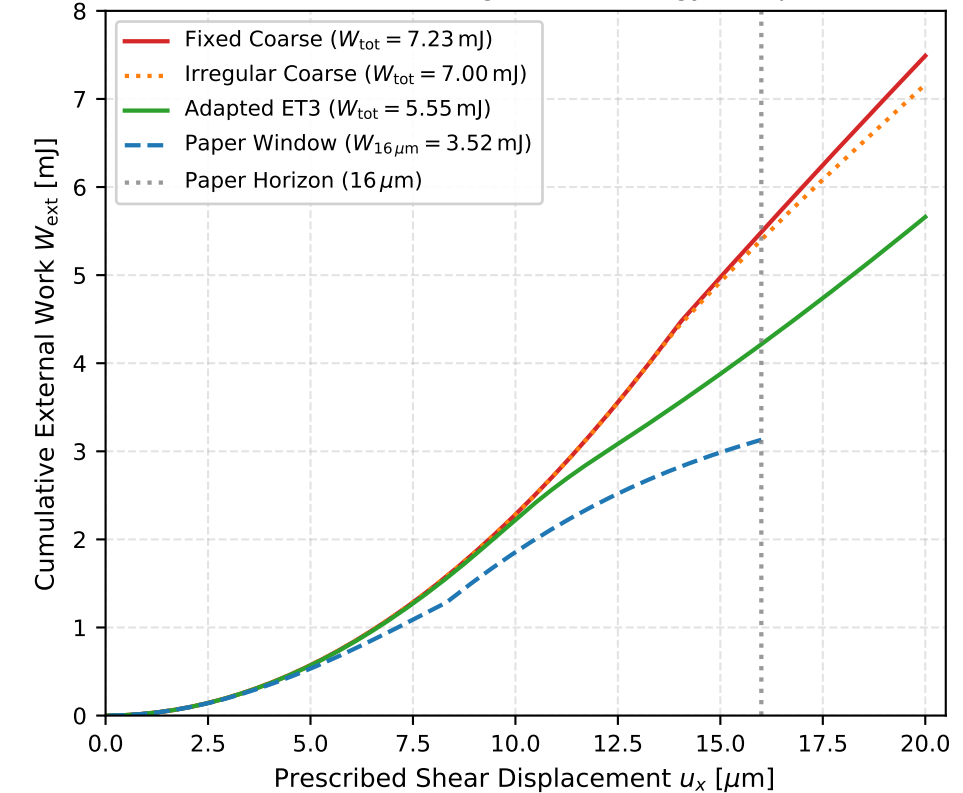
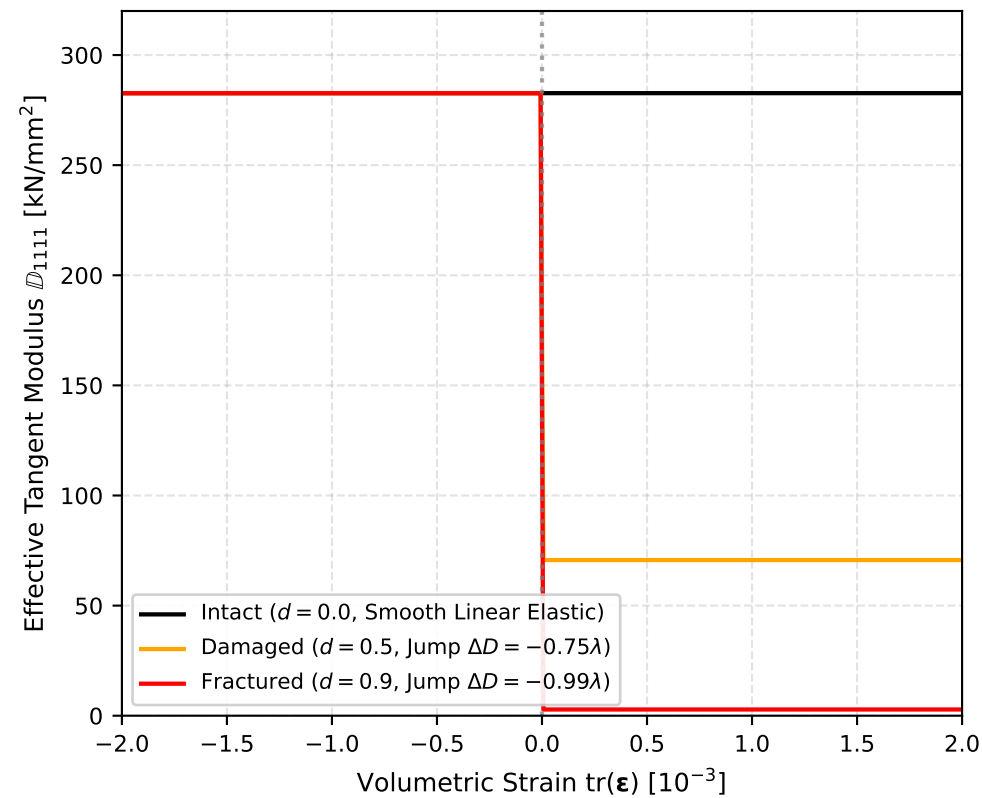


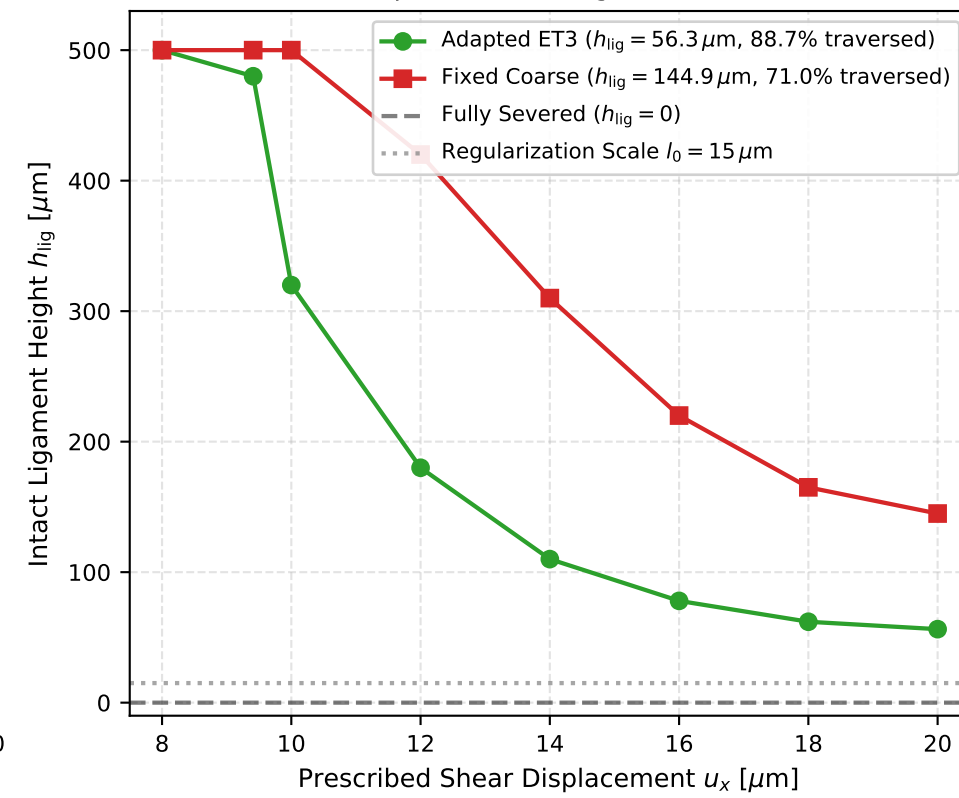
(a) Mode-II Global Load-Displacement Response

(b) Initial Elastic Stiffness Parity ($R^2 > 0.9999999$)

(c) External Work Integration & Energy Dissipation

(d) Miehe Spectral Split: Subgradient Jump at $\text{tr}(\boldsymbol{\epsilon}) = 0$ 

(e) Crack Tip Advance & Ligament Penetration

(f) Multi-Field Adaptive Controller: $\eta_K = \max(\eta_\sigma, \eta_d)$ 